Muhammad Soban Zamir

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EDUCATION

National University of Sciences & Technology, Islamabad Sep 2024 - June 2028
BSc Electrical Engineering

- Relevant coursework: Microprocessor Systems, Machine Learning, Electronic Circuit Design, Data Structures & Algorithms
- Dean's List Student (**CGPA:** 3.60/4.00)

PROFESSIONAL EXPERIENCE

National Centre of Artificial Intelligence **Undergraduate Research Intern** June 2025 – Sep 2025

- Selected through a competitive programming test and a detailed interview for an internship at TUKL – NUST R&D Centre and Deep Learning Lab, National Centre of Artificial Intelligence, (NCAI)
- Worked on a deep learning pipeline for dental X-ray classification. Redesigned the clinical text embedding to Improve representation quality and clustering space.
- The new algorithm improved the active learning loop for data labelling, which significantly enhanced classification performance.
- Trained and optimized custom CNNs (89% accuracy) and Transformer-based sequence models (93% accuracy) using Python and PyTorch.
- Earned advanced certifications from Stanford University, DeepLearning.AI, and IBM, all scored 95+

TECHNICAL SKILLS

Engineering Tools: MATLAB, AutoCAD, MultiSIM, Proteus, Atmel Studio, Pspice
Programming Languages: Python, C, C++, MATLAB, HTML, Assembly
Deep Learning: TensorFlow, PyTorch, Jupyter, NumPy
Computer Skills: Microsoft Office 365

PROJETCS

Discrete Logic Based Electronic Slot Machine Nov 2025 - Dec 2025

- Designed and implemented a discrete logic electronic slot machine, architected to emulate commercial casino machine circuitry.
- Implementation involved using asynchronous random counters made from latches and flip flops, logic gates, and seven segment displays to generate and display random outcomes.
- Developed a magnitude comparator using XOR logic circuitry to detect winning conditions and signal winning LEDs to turn on

NLP Grammar Correction System Using Transformers Oct 2025 - Nov 2025

- Developed an interactive NLP-based grammar correction system using Transformer models for real-time text refinement
- Implemented using Python, PyTorch, and Transformer models (BERT/T5), deployed via Gradio on Google Colab
- Part of a larger pipeline involving sentence correction, audio to text conversion and language translation

MOSFET Audio Amplifier Oct 2025 - Nov 2025

- Designed and implemented a MOSFET based audio amplifier capable of amplifying low level AC signals.
- The design was simulated in PSPICE, followed by hardware implementation and performance validation, addressing practical considerations such as noise, gain control, and load-driving capability.
- The circuit utilized an IRF9540 MOSFET with carefully designed gate biasing, source degeneration, and AC coupling to ensure stable operation, low distortion, and reliable signal amplification

CERTIFICATIONS

- **Higher Education Commission (HEC):** IoT for Everyone
- **International Business Machines:** LLM & Generative AI Series
- **DeepLearning.AI**: Deep Learning Specialization
- **Stanford Online:** Machine Learning Specialization